Oscar Stephen Yasunaga

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EDUCATION

University of Toronto, St. George Campus **2023 - *2027***

*BSc. Physics Major, Math and Stats Minor - **c**GPA: 3.7/4.0*

Relevant Coursework: Neural Networks and Deep Learning (Joint Graduate), Computational Methods in Physics (Joint Graduate), Introduction to Machine Learning, Probability, DSA, MV Calc & Lin Alg with Proofs.
Awards: UofT Entrance Award ($12k), ThinkingMachine ($5k), 4x Dean's List Scholar, TAISI Fellowship.
Achievements: QSITE Hackathon 24 - Top 2 out of 364, DeerHacks25 - Top 3 out of 128,

AI SAFETY EXPERIENCE

Technical Alignment Research Accelerator (TARA) **Remote**

Teacher *Aug 2026 - Present*

- Teach the ARENA curriculum to 58+ students across TARA's Australian cluster as part of a 14-week, part-time technical AI safety program, earning a 4/5 average rating from participants
- Facilitate weekly end of day sessions on timelines, fellowships, research landscape, and fieldbuilding.

Toronto AI Safety Initiative (TAISI) **Toronto, Canada**

Teacher/Facilitator *May 2026 - Present*

- Developed ARENA style tutorial notebook to replicate Emergent Misalignment, inducing broad misalignment via SFT fine tuning on the bad aesthetics preferences dataset [Colab]
- Taught 30+ undergraduate and graduate students across topics in EM, mechanistic interpretability, and evaluation science, scoring an average learner and program ratings of 4.5/5.
- Served as a facilitator talking about instrumental convergence, and threat model designs to 10 students.

Supervised Program for Alignment Research (SPAR) **Remote**

Supervisors: Nischal Mainali (PIBBS Researcher) *Sep 2026 - Present*

- Developing theory and methods (e.g., temporal sparse dictionaries, switching linear dynamical systems) to decompose LLM training trajectories into sparse, recurring learning events and to model how those events compose.

AI Safety Research Assistant, JinesisAI Lab **Toronto, Canada**

Supervisors: Prof Zhijing Jin, Terry Jingchen Zhang, Florent Draye *Jan 2026 - Present*

- **Developing a robust benchmark framework** for sycophancy grounded in social psychology principles.
- Developing SycoScope which is a comprehensive framework for mechanistic interpretability detectors of sycophancy.

AI Safety Research **Toronto, Canada**

Supervisors: Cole Blondin, Vasu Sharma *March 2025 - July 2025*

- **Co-author of ACL SRW paper**, extending a concept cone framework to propositional truth activations.
- **Demonstrated cone-based representations:** Larger scale models Qwen2.5-14B, and Gemma2-9B causally encode at least 5-orthogonal vectors that represent truth with a 100% answer switch rate.
- Through PCA projections of residual stream activations across layers, showed linear separability of truth.

PUBLICATIONS

SuperSycophantic: Stress-Testing Frontier Models from Single- to Multi-Turn Sycophancy

Terry Jingchen Zhang, **Oscar Y**, Wenyuan Jiang, Jessica Bo, Florent Draye, Aydin Javadov, Arth Singh, Lucille Muir, Victoria Oldemburgo De Mello, Bernhard Scholkopf, Zhijing Jin
Accepted at **NeurIPS 2026**

RLViz: An Interactive Visualization to Aid Student Understanding of Bellman Equations

Oscar Y, Dylan Ma, Chandra Gummaluru
Submitting to **AAAI Symposium Educational Advances in Artificial Intelligence. 2026 (In Preparation)**

GOES-ABI Data Cube: Eight Years of Unified Geostationary Satellite Observations
Mickell Als*, Misha Schwartz*, Chen Yu-An, **Oscar Y**, Benjamin Lee, Steve Easterbrook
Accepted at **Climate Informatics 2026**

From Directions to Cones: Exploring Multidimensional Representations of Propositional Facts
Stanley Yu*, Vaidehi Bulusu*, **Oscar Y**, Clayton Lau, Cole Blondin, Sean O'Brien, Kevin Zhu, Vasu Sharma
Accepted at **ACL Student Workshop 2025**

RESEARCH EXPERIENCE

AI Generalization Directed Reading Program **Toronto, Canada**
Supervisors: Haoming Meng *April 2026 - Aug 2026*

- Read literature on neural collapse, inductive biases, grokking, lazy vs active learning to build background on generalization phenomena exhibited in trained neural networks (Based on this grad course)
- Investigating Jacobian-based explanations of generalizations, focusing on eigendecomposition and non-normality of cross-token and per-token Jacobians at each layer

RL Pedagogy Research **Toronto, Canada**
Supervisor: Prof Chandra Gummaluru *Dec 2025 - Present*

- **Implementing an educational web-based MDP visualizer** to supplement ROB311 (Reinforcement Learning), by animating Value Iteration process, the Bellman Equation recursion proof.
- Supports both time in/dependent policies, exact policy evaluations, POMDP and model unknown setups.

Climate Physics AI Research **Toronto, Canada**
Supervisors: Mickell Als, Prof Steve Easterbrook *October 2025 - April 2026*

- **Co-author of Climate Informatics paper**, developing a pipeline that reorganized 8 years of GOES Advanced Baseline Imager Full Disk observations into Zarr data cubes, enabling large scale modelling.
- Constructed quality assurance tests of pipeline identifying **6.7% of all temporal scans** were corrupted.

SERVICE

- Teaching Assistant Toronto AI Safety Initiative Summer Intensive (TAISI) 2026
- Reviewer for the Neural Information Processing Systems (NeurIPS) NewinML Workshop (2026)
- Reviewer for the Neural Information Processing Systems (NeurIPS) AI4Good Workshop (2026)
- Reviewer for the International Conference on Machine Learning (ICML) AI4Good Workshop (2026)
- UTMIST Research Manager for the Geometry of Sycophancy Project (2025)
- Mentor for the Physics Student Union's early undergraduate program | PHYSU (2025)

Additional Information

Coding Languages: Proficient: Python, Java, R; Working knowledge: SQL, MATLAB, Javascript, HTML, CSS
Languages: English and Chinese
Skills: NumPy, SciPy, Pandas, OpenCV, Scikit-Learn, PyTorch, NNSight, Unsloth, Trl, TransformerLens, Scikit-Learn, data visualization, web scraping, Tinker, WanDB, HPC, SLURM, deep learning, alignment, mechanistic interpretability, scalable oversight, robustness, jailbreaks
Hobbies: Standup Comedy, Improv Comedy, Chess